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DATA STRUCTURE AND ALGORITHM (LAB EXERCISES AND PRACTICE)

OUTPUT TRACING QUESTION

PSEUDO CODE (TRACING)

Based on the following pseudo code in **Figure 1**, complete the trace table given in **Table 1**.

[5 marks]

```

1. START
2. READ n, m
3. IF (n >= m)
    3.1 START_IF
        3.1.1 IF (n > 10)
            3.1.1.1 START_IF
                3.1.1.1.1 IF (m > 10)
                    3.1.1.1.1.1 START_IF
                        3.1.1.1.1.1.1 PRINT "both n and m is greater than 10"
                    3.1.1.1.1.1.2 END_IF
                3.1.1.1.1.2 IF (n == m)
                    3.1.1.1.1.2.1 START_IF
                        3.1.1.1.1.2.1.1.1 PRINT "n is equal to m"
                    3.1.1.1.1.2.2 END_IF
            3.1.1.2 END_IF
        3.2 END_IF
4. ELSE
    4.1 PRINT (n-m)*2
5. PRINT n, m
6. END

```

Figure 1

ANSWER**Table 1**

n	m	Output
0	0	0 0
10	0	10 0
20	10	20 10
20	20	both n and m is greater than 10 n is equal to m 20 20
0	10	-20 0 10

PSEUDO CODE (TRACING)

Based on the following pseudo code in **Figure 2**, complete the trace table given in **Table 2**.

[9 marks]

Data to be used for this problem is:

noItems

3

itemID CP SP units

1111 32 40 10

2222 25 30 5

3333 57 65 2

```

1. START
2. SET Total to 0
3. READ noItems
4. WHILE (noItems IS NOT 0)
    4.1 START_WHILE
        4.1.1 READ itemID, CP, SP, units
        4.1.2 P = (SP * units) - (CP * units)
        4.1.3 ADD P to Total
        4.1.4 PRINT itemID, P
        4.1.5 SUBTRACT 1 from noItems
    4.2 END_WHILE
5. PRINT Total
6. END

```

Figure 2

ANSWER:

Table 2

Total	noItems	P	Output	
0	3	80	1111	80
80	2	25	2222	25
105	1	16	3333	16
121	0		121	

FLOWCHART (TRACING)

Use the flowchart in **Figure 3** to answer Question 3.

[10 marks]

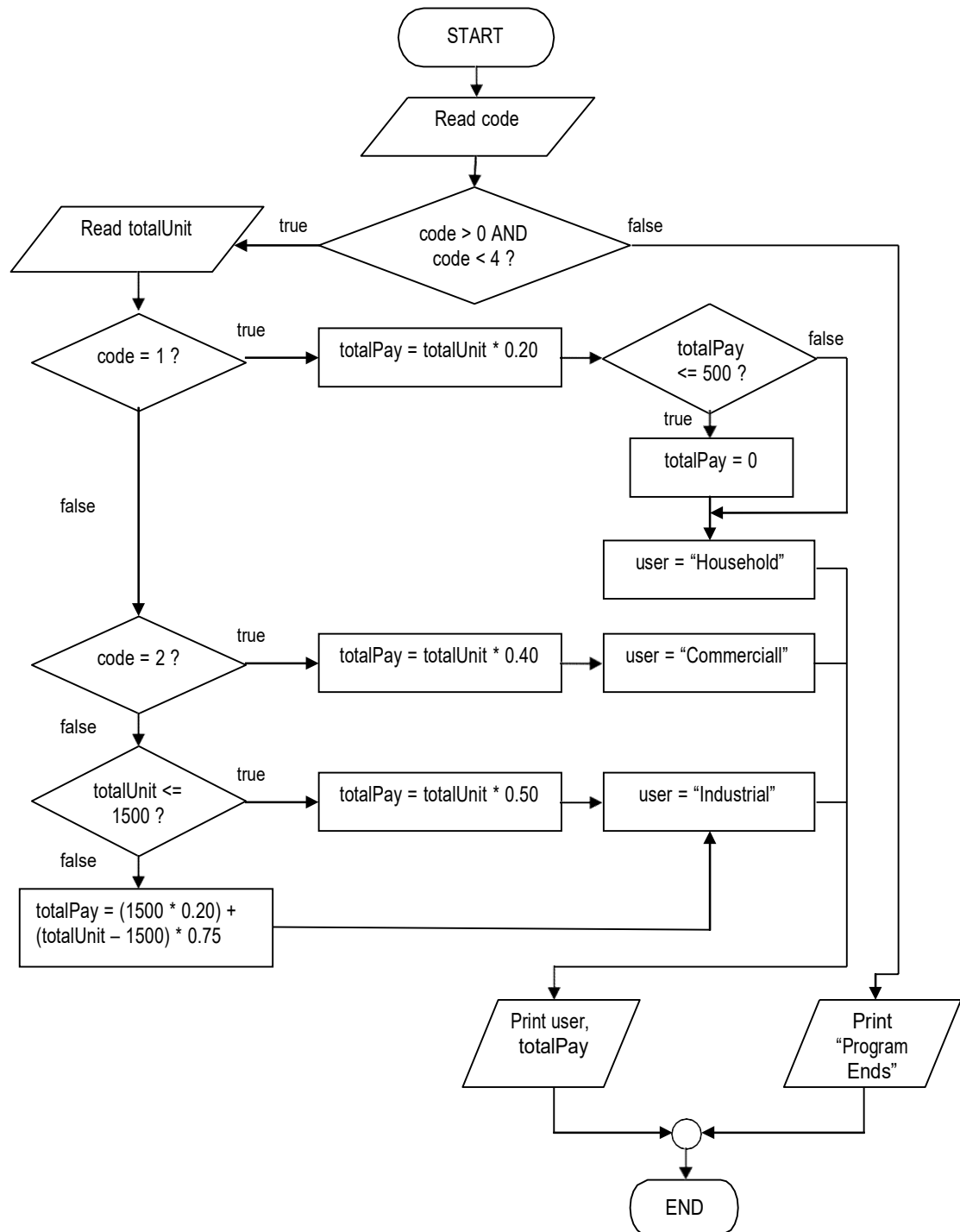


Figure 3

Trace the variables of the flowchart in **Figure 3** using the following input values:

	<u>Code</u>	<u>totalUnit</u>
i.	0	200
ii.	1	375
iii.	2	550
iv.	3	4500
v.	1	30
vi.	2	1000
vii.	3	1200

ANSWER:

Fill your answers in the **Table 3** provided below.

Table 3

code	totalPay	Output
0		Program Ends
1	75	Household 0
2	220	Commerciall 220
3	2550	Industrial 2550
1	6	Household 0
2	400	Commerciall 400
3	600	Industrial 600

FLOWCHART (TRACING)

Trace the flowchart below in **Figure 4** by fill in the table given in **Table 4**. [15 marks]

!

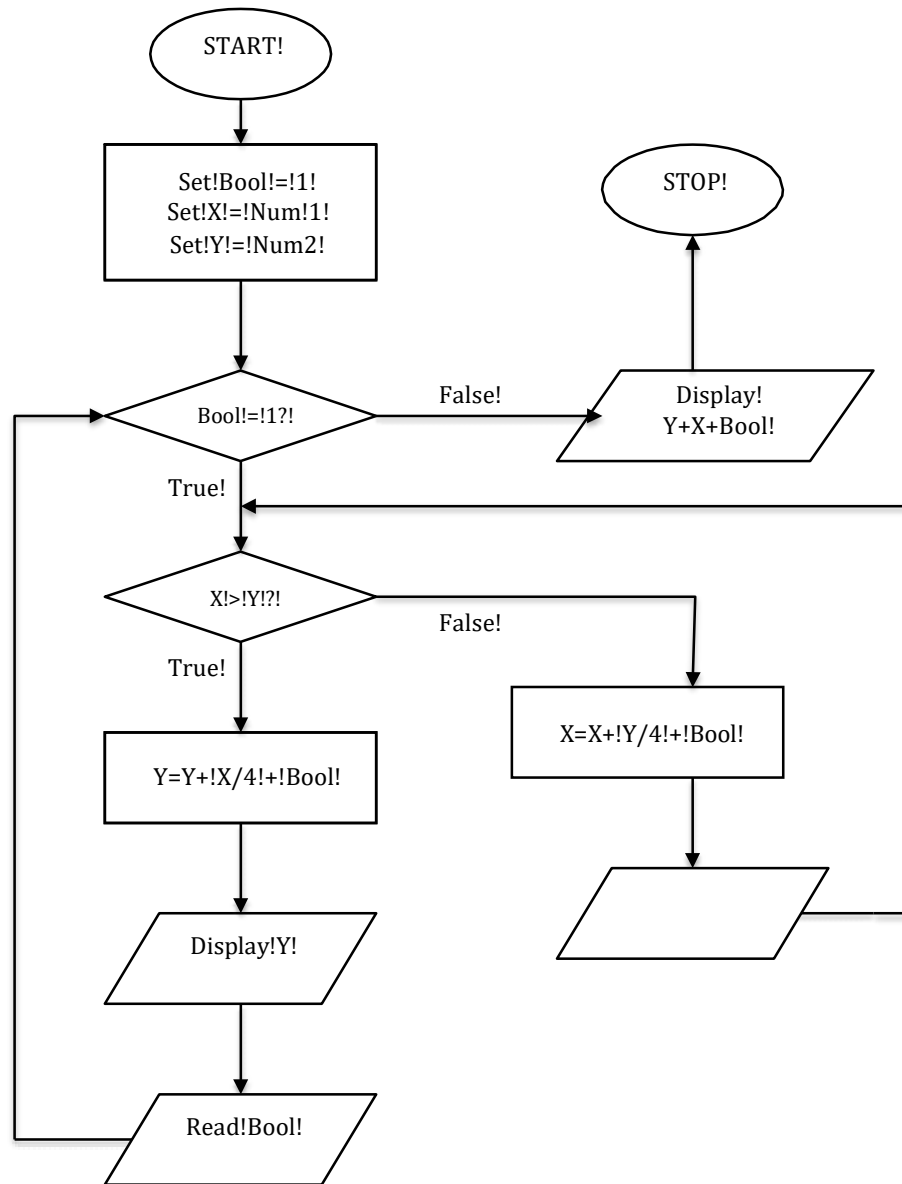


Figure 4

ANSWER:**Table 4**

Num1	Num2	Bool	X	Y	Output
7	8	1 1 2			
23	5	1 1 -5			
32	23	1 1 1 10			

PROBLEM SOLVING QUESTION***PROVIDE COMPLETE PSEUDO CODE*****[20 marks]**

Write a pseudo code for a program that will implement the following decision table in **Table 5**. The program will print the input grade point and the class of degree based on a user input. The program will terminate the loop when a user input a sentinel value other than 'y' or 'Y'.

Table 5

GRADE POINT	Class of Degree
0.0 – 0.99	Failed
1.0 – 2.00	General degree
2.1 – 2.7	Second class lower
2.71 – 3.69	Second class upper
3.7 – 4.00	First Class

ANSWER:

1. START
2. DO
 - 2.1. READ Gradepoint
 - 2.2. IF (Gradepoint>=0.00 && Gradepoint<=4.00)
 - 2.2.1. START_IF
 - 2.2.1.1. IF (Gradepoint>=0.00 && Gradepoint <1.00)
 - 2.2.1.1.1. Print "Failed"
 - 2.2.1.2. ELSE_IF (Gradepoint>=1.00 && Gradepoint <2.10)
 - 2.2.1.2.1. Print "General degree"
 - 2.2.1.3. ELSE_IF (Gradepoint>=2.10 && Gradepoint <2.71)
 - 2.2.1.3.1. Print "Second class lower"
 - 2.2.1.3. ELSE_IF (Gradepoint>=2.71 && Gradepoint <3.70)
 - 2.2.1.3.1. Print "Second class upper"
 - 2.2.1.3. ELSE_IF (Gradepoint>=3.70 && Gradepoint <=4.00)
 - 2.2.1.3.1. Print "First Class"
 - 2.2.2. END_IF
 3. Print "Do you want to enter another grade point? (y/Y to continue)"
 4. READ userinput
 5. WHILE (userinput == 'y' || userinput == 'Y')
 6. END

PROVIDE COMPLETE PROGRAM**[30 marks]**

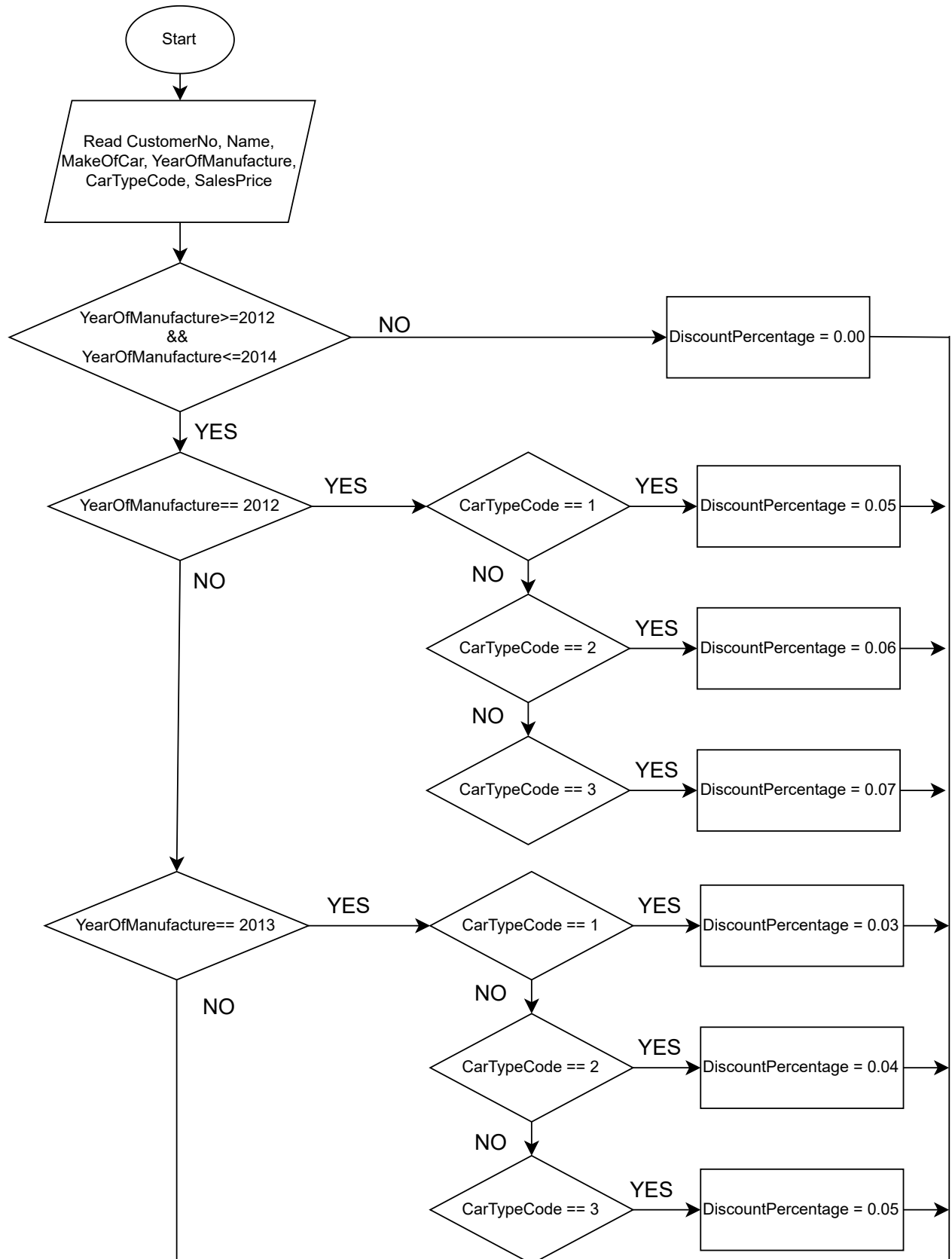
Mode Auto Dealer is a used car business owned by Mr. Muhammad. He employs you to develop a program that will calculate the sales discount to be applied to a vehicle, based on its year of manufacture and type. The discount is extracted from a two-dimensional table as in **Table 6** below:

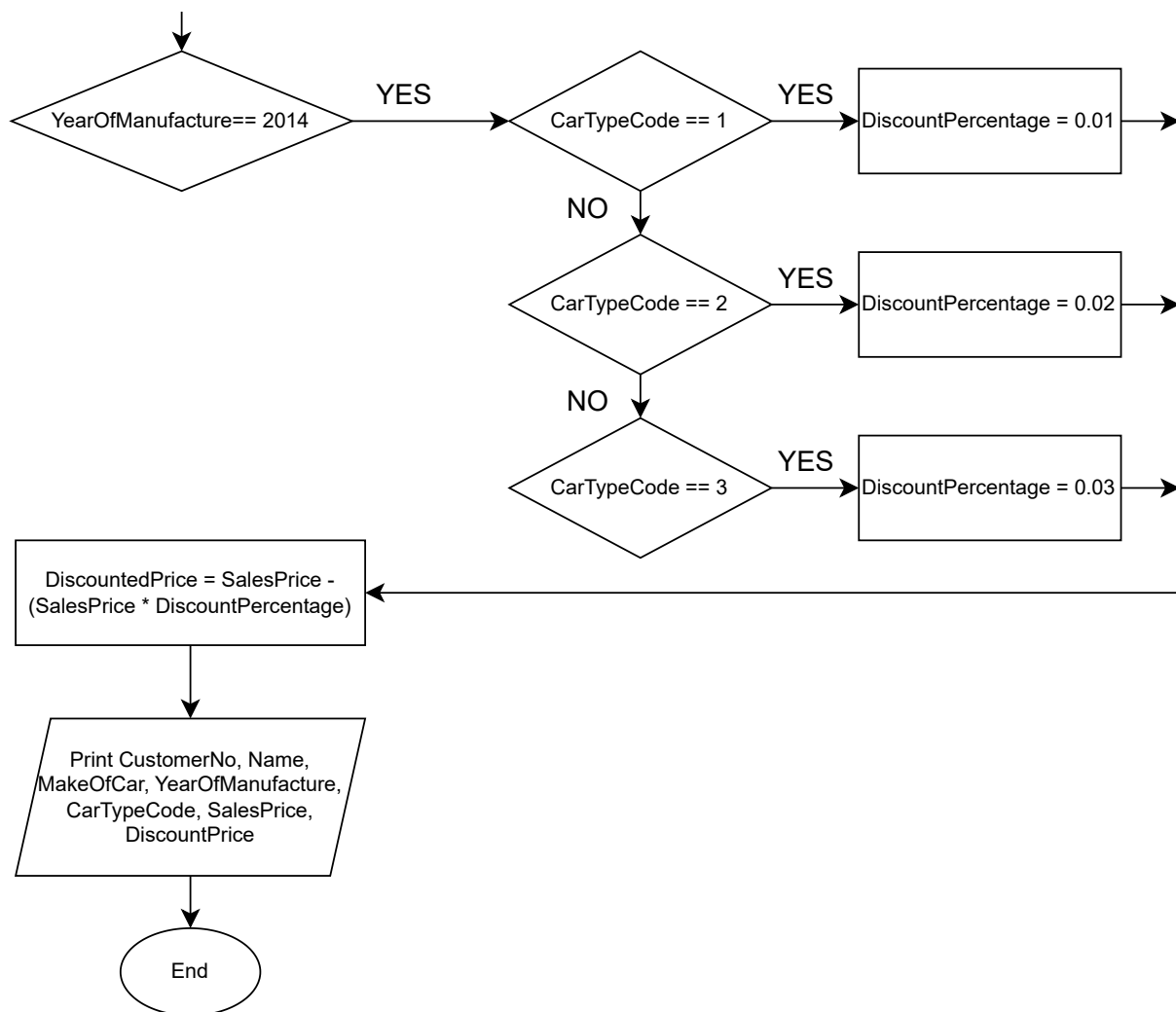
Table 6

Year of manufacture	Discount Percentage		
	Small	Medium	Luxury
	1	2	3
2012	0.05	0.06	0.07
2013	0.03	0.04	0.05
2014	0.01	0.02	0.03

The year of manufacture of the vehicle is divided into three categories (2012, 2013 and 2014), and the type of car is divided into three categories (small, medium and luxury). No discount is given for a vehicle older than 2012. Your program is to read the vehicle file, which contains the customer number and name, the make of car, year of manufacture, car type code (1, 2, or 3) and the sales price. Use the year of manufacture and the car type code as guidelines to retrieve the discount percentage table, then apply the discount percentage to the sales price to determine the discounted price of the vehicle. Print all the details including discounted price. Before writing the program you are required to plan your problem solving steps in a flowchart(s).

Note: Additional papers will be given to answer this question.

ANSWER



```
1 #include <iostream>
2 #include <iomanip>
3 #include <string>
4 using namespace std;
5
6 int main()
7 {
8     int customerNo;
9     string customerName, makeOfCar;
10    int yearOfManufacture, typeCode;
11    double salesPrice, discountPercentage = 0.0, discountPrice;
12
13    cout << "Enter Customer Number: ";
14    cin >> customerNo;
15    cout << "Enter Customer Name: ";
16    cin.ignore();
17    getline(cin, customerName);
18    cout << "Enter Make of Car: ";
19    getline(cin, makeOfCar);
20    cout << "Enter Year of Manufacture: ";
21    cin >> yearOfManufacture;
22    cout << "Enter Type of Car (1:Small, 2:Medium 3 Luxury): ";
23    cin >> typeCode;
24    cout << "Enter Sales Price: ";
25    cin >> salesPrice;
26
27    if (yearOfManufacture >= 2012 && yearOfManufacture <= 2014)
28    {
29        if (yearOfManufacture == 2012)
30        {
31            if(typeCode ==1) discountPercentage = 0.05;
32            else if(typeCode ==2) discountPercentage = 0.06;
33            else if(typeCode ==3) discountPercentage = 0.07;
34        }
35        else if (yearOfManufacture == 2013)
36        {
```

```

31         if(typeCode ==1) discountPercentage = 0.05;
32         else if(typeCode ==2) discountPercentage = 0.06;
33         else if(typeCode ==3) discountPercentage = 0.07;
34     }
35     else if (yearOfManufacture == 2013)
36     {
37         if(typeCode ==1) discountPercentage = 0.03;
38         else if(typeCode ==2) discountPercentage = 0.04;
39         else if(typeCode ==3) discountPercentage = 0.05;
40     }
41     else if (yearOfManufacture == 2014)
42     {
43         if(typeCode ==1) discountPercentage = 0.01;
44         else if(typeCode ==2) discountPercentage = 0.02;
45         else if(typeCode ==3) discountPercentage = 0.03;
46     }
47 }
48 else {discountPercentage = 0.0; }
49
50 discountPrice = salesPrice - (salesPrice * discountPercentage);
51
52 cout << fixed << setprecision (2);
53 cout << "\n --- Sales Discount Information --- \n";
54 cout << "Customer Number: " << customerNo << endl;
55 cout << "Customer Name: " << customerName << endl;
56 cout << "Make of Car : " << makeOfCar << endl;
57 cout << "Year of Manufacture : " << yearOfManufacture << endl;
58 cout << "Type of Car: " << (typeCode == 1 ? "Small" : typeCode == 2 ? "Medium" : "Luxury") << endl;
59 cout << "Sales Price: $" << salesPrice << endl;
60 cout << "Discount Percentage: " << (discountPercentage * 100) << "%" << endl;
61 cout << "Discounted Price: $" << discountPrice << endl;
62
63 return 0;
64
65 }

```



C:\Users\user\Downloads\DS/ X + v

```
Enter Customer Number: 25
Enter Customer Name: Koo Xuan
Enter Make of Car: Range Rover
Enter Year of Manufacture: 2013
Enter Type of Car (1:Small, 2:Medium, 3:Luxury): 3
Enter Sales Price: 100000
```

--- Sales Discount Information ---

```
Customer Number: 25
Customer Name: Koo Xuan
Make of Car : Range Rover
Year of Manufacture : 2013
Type of Car: Luxury
Sales Price: $100000.00
Discount Percentage: 5.00%
Discounted Price: $95000.00
```

Process exited after 40.91 seconds with return value 0
Press any key to continue . . . |